

Aerial Display Method Using a Flying Screen with an IR Marker and Long Range Dynamic Projection Mapping

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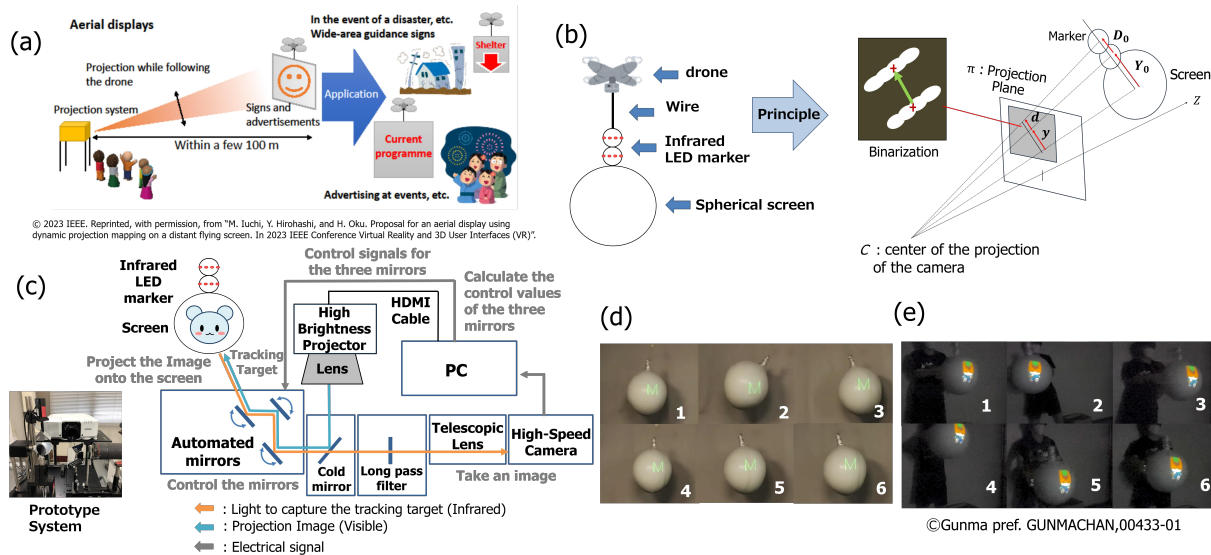


Figure 1: (a) The final goal of this study. (b) Spherical screen and tracking marker using two infrared LEDs to enable the estimation of the screen's center. (c) System configuration of the long range projection equipment. (d) Sequential photographs of the projection experiment of laser patterns onto the screen with an IR marker suspended from a drone. (e) Sequential photographs of the dynamic projection mapping using the long range projection equipment.

ABSTRACT

Our group proposed a method of aerial display using a flying screen suspended from a drone and dynamic projection mapping on the screen. In this study, we propose two methods as extensions of this previous study. First, we propose a structure of a large screen with a specially designed LED marker. The proposed screen structure is suitable for suspending from a drone. The LED marker has a special structure that allows the screen center position to be estimated from the captured image. Successful and stable projection of laser patterns onto the screen prototype suspended from a flying drone was demonstrated. In this experiment the distance between the screen and the projection system was approximately 16 m. Second, we propose a method that enables long range dynamic projection mapping using a high-brightness projector. Stable projection of

an animation of a 2D character on a manually moving screen approximately 16 meters away from the projection system was also successfully demonstrated.

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1 INTRODUCTION

In recent years, technologies that enable visual representation of images in the sky, such as drone shows, have been gaining attention. However, drone shows require the use of hundreds of drones equipped with lights, resulting in significant costs and making it difficult to utilize them for small-scale events.

Iuchi et al. have reported an aerial display method based on dynamic projection mapping onto a screen suspended from a single drone [Iuchi et al. 2023]. The reported projection equipment consisted of the "Telescopic three-mirror gaze control system" [Ogasawara et al. 2018], which rapidly controls the camera's gaze by adjusting the angles of three mirrors placed in front of a telescopic camera,

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and the "Laser Display" [Iuchi et al. 2023], which controls two rotating mirrors to project laser patterns. Tracking was performed using infrared LEDs attached to the target as a marker, and dynamic projection mapping onto distant objects was achieved by projecting laser patterns in the camera's gaze direction.

However, the proposed method in the study has several problems. In the experiment by Iuchi et al., the screen size was small (approximately 30 cm square), and a larger screen is needed for the application as an aerial display. However, as the screen size increases, directly attaching a large screen to the bottom of the drone, as in the previous study, would obstruct takeoff and landing. To avoid this problem, one solution is suspending the screen with long wires. However our preliminary experiments showed that this structure made the flat screen unstable against the airflow generated by the drone's propellers. Additionally, there was another problem that the projected image protrudes when the screen was tilted because the projection did not account for the screen's tilt.

In this study, we propose (i) the structure of a large screen suitable for suspending from a drone, (ii) a marker for stable projection on the screen by estimating the screen tilt, and (iii) a long range dynamic projection mapping method by replacing the laser display used in the previous study with a high-brightness projector with a telescopic projection lens. A prototype screen and projection system was developed and demonstrated the validity of the proposed methods.

2 PROPOSAL FOR A SPHERICAL SCREEN

In this study, we propose a spherical screen as shown in Figure 1(b). The jet flow blowing onto the sphere is bent along the surface of the sphere due to a phenomenon known as the Coandă effect [Reynolds 1974]. In this case, the sphere is drawn towards the airflow due to the reactive force exerted by the fluid. With this property, by utilizing the airflow generated by the drone's propellers blowing downward, it becomes possible to keep the spherical screen in place directly beneath the drone. Even if there is a certain distance between the screen and the drone, it can be stably suspended.

3 ESTIMATION METHOD FOR THE CENTER OF THE SCREEN USING AN IR MARKER

Next, we propose a method to estimate the center of the screen using tape-like infrared LEDs arranged at two locations as markers for tracking. In this case, we have developed markers with a structure that can be attached to the edge of the screen, as attaching markers directly to the screen would obstruct the projection. By analyzing the position relationship of the two infrared LEDs in the images captured by the camera, the center of the screen can be estimated. The outline of this method is shown in Figure 1(b). In this context, D_0 represents the vector connecting the infrared LEDs of the markers in 3D space, Y_0 represents the vector connecting the screen's center to the markers in 3D space, d represents the vector connecting the centroids of the two infrared lights appearing in the camera image, and y represents the vector connecting the screen's center position on the projection plane to the markers.

Here, we assume that the screen is in the far distance from the camera, so weak perspective projection is assumed. Since the ratio $k = \|D_0\|/\|Y_0\|$ can be obtained from the known vectors D_0 and

Y_0 , the vector y on the projection plane using weak perspective projection can be expressed using the vector d on the camera image as shown in Equation (1).

$$y = \frac{1}{k}d = \frac{\|Y_0\|}{\|D_0\|}d \quad (1)$$

Once the vector y is obtained, the position of the screen center in the image can be determined, and the direction to project the image onto the screen center can be estimated, enabling projection onto the screen center.

4 PROPOSAL FOR THE LONG RANGE DYNAMIC PROJECTION MAPPING METHOD

We propose a system that enables long-range projection mapping by replacing the laser display, which was integrated into the projection equipment developed in the research by Iuchi et al., with a high-brightness projector equipped with a telescopic projection lens. The system configuration of the system is shown in Figure 1(c).

This system allows projection mapping onto a dynamic object in the far distance captured by the camera by reflecting the high brightness projector's image onto three mirrors that control the direction of the telescopic camera's gaze.

Furthermore, the projection images appear tilted from the perspective of the observer due to the nature of the telescopic three-mirror gaze control system. To correct this rotation effect, previously rotated image was projected according to the algorithm by Higuchi et al. [Higuchi and Oku 2021].

5 EXPERIMENTS

To demonstrate the effectiveness of the proposed method, we conducted evaluation experiments. The results of the two projection experiments are shown in Figure 1(d) and Figure 1(e). From these experiments, we confirmed that stable projections were achieved even when the screen was tilted.

6 CONCLUSION

In this paper, (i) the structure of a large screen, (ii) a marker for stable projection and (iii) a use of high-brightness projector were proposed for projection based aerial displays.

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